LA GRANDEE INTERNATIONAL COLLEGE

Simalchaur, Pokhara Nepal

A Final Report

On

"Rock, Paper, Scissor"

Submitted to:

Bachelor of Computer Application (BCA) Program

In partial fulfilment of the requirements for the degree of BCA under

Pokhara University

Submitted by:

Name:	Course	Semester	P.U. Registration Number
Divya Sharma	BCA	2nd	2025-1-53-0323
Sabina Puwanli	BCA	2nd	2025-1-53-0341

Date: 12/07/2026

Acknowledgement

The satisfaction that accompanies after the successful completion of any task will be incomplete without mentioning the people whose ceaseless and relentless cooperation, constant guidance and encouragement made this project possible.

We are grateful to our project supervisor **Mr Arpan Pokhrel**, our faculty teacher **Mr. Ramesh Chalise,** for the guidance, inspiration and constructive suggestions that helped us in the preparation of this project.

We are also appreciative among each other and have understood that teamwork, the designation of the task per the skillset one portrays, constant synchronisation and monitoring of progress and instilling new knowledge and skill is imperative for the success of any given work.

Sincerely,

Divya Sharma

Sabina Puwanli

Student Declaration

We hereby declare that we are the only authors of this work and that no sources other than the listed here have been used in this work. I further confirm that the project work "Rock, Paper, Scissor" and hence this report is my original work and has not formed the basis for the award of any other degree or other similar titles.

Name of the Students	Symbol Number
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BCA, 2^{nd} semester

Date: 12/07/2026

Supervisor Acceptance

I, hereby, declare that the project entitled "Rock, Paper, and Scissor" is done under my supervision by Sabina Puwanli and Divya Sharma during their 2nd Semester in partial fulfilment of the requirements for the degree of Bachelor of Computer Application under Pokhara University is completed to my satisfaction and I recommend to be processed for final evaluation.

Arpan Pokhrel, Lecturer
Date: 12/07/2026

Letter of Approval

We certify that we have examined this report entitled "**Rock, Paper, and Scissor**" and are satisfied with the project defence. In our opinion it is satisfactory in the scope and qualify as project in partial fulfilment of the requirements for the degree of **Bachelor of Computer Application (BCA)** under **Pokhara University.**

Program Coordinator	**Examiner**	**Principal**
Ankit Poudyal	Er. Dikshan Gurung	Er. Kiran KC

Date: 12/07/2026

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Abbreviations

Abbreviation	Full Form
AI	Artificial Intelligence
C	C Programming Language
CRUD	Create, Read, Update, Delete
DFD	Data Flow Diagram
PU	Pokhara University
RTM	Requirement Traceability Matrix
SDLC	Software Development Life Cycle
STLC	Software Testing Life Cycle
TC	Test Case
UAT	User Acceptance Testing

1 . Introduction

Rock, Paper, Scissors is one of the most popular and oldest hand games played throughout the world. The game is simple, entertaining and commonly used for decision making purposes. Traditionally, the game is played between two players where each player chooses whether rock, paper, or scissor. In this computerized version, the user plays against computer and the winner is determined according to the rules: rock defeats scissors, scissors defeat paper, and paper defeats rocks. (Balagurusamy, 2020)

This project focuses on developing an advanced Rock Paper Scissors game using C programming language with several advanced features such as login system, registration system, CRUD operations, score management, multiplayer gameplay, account deletion, leader board system, match history, and file handling. The project is designed to simulate a real real-world gaming system where users can create accounts, store scores permanently, compete with other players, and manage their data efficiently. (Kernighan & Ritchie, 1988; Prata, 2014)

The system will allow users to create accounts and securely log in using usernames. User data is stored permanently using file handling techniques. Once logged in, players can play against the computer or compete with another player in multiplayer mode. Scores are recorded and stored in files so that player performance can be tracked even after the program is closed. (Prata, 2014)

The implementation of CRUD operation is one of the most important highlights of the project. CRUD stands for Create, Read, Update, and Delete. In this project, users can create new accounts, read player details and scores, update records automatically through gameplay, and delete accounts permanently. (Microsoft, n.d.)

File handling is another important component of the project. Since the C language does not include a built-in data base management system, text files are used to store player records, scores, and match history. The multiplayer feature increases interaction and competitiveness between users. Instead of limiting gameplay to computer-based interaction only, the system allows two users to compete directly against each other. This improves our experience and makes the game more engaging.

This project also includes additional innovative feature such as tournament mode and streak bonus system. Tournament mode allows users to compete in multiple rounds continuously, while the streak bonus system reward users for consecutive wins. These features increase excitement and provide better gaming experience.

The project demonstrates the practical application of various programming concepts including loops, conditional statements, arrays, structures, functions, file handling and user authentication which helps beginner programmers understand how multiple concepts can be combined together to create a complete software application. (Balagurusamy, 2020; Tutorials Point, n.d.)

Overall, the advanced Rock Paper Scissors game is an educational and interactive project that combines entertainment with software development knowledge. It provides practical understanding of game logic, user management systems, and permanent data storage through file handling.

2. Problem Statement

The traditional Rock, Paper, and Scissor game has several limitations that reduce its usability and functionality. The following problems motivated the development of this project:

1. No Permanent Data Storage
Game results and player statistics are not stored permanently. Once the game ends, all scores and records are lost.

2. Absence of Match History and Leaderboard
Players cannot review previously played matches or compare their performance with other players through a leaderboard.

3. Limited Gameplay Features
The traditional game offers only basic gameplay and lacks additional features such as multiplayer support, player profiles, and organized game management, reducing the overall user experience.

4. Lack of User Management

The traditional game does not provide a registration or login system, making it impossible to identify different players or maintain individual player records

To overcome these Limitations, this project develops a console-based Rock Paper Scissors Game using C programming language with user registration, login, single player and multiplayer modes, score management, match history, leaderboard, and file handling for permanent data storage.

3 .Objective

The main objective of this project is to develop a complete Rock Paper Scissors gaming system using C programming language. Here are some specific objectives listed below:

1. To implement multiple gameplay modes including player versus computer and multiplayer functionality.
2. To use file handling for storing player records, maintaining match history, and displaying scoreboards.
3. To create Dashboard that provides easy access to gameplay, match history, leader board and user options.

4 .Background Study

The evolution of game development from traditional board and hand games to digital applications has significantly enhanced user engagement and accessibility. Rock-Paper-Scissors, being one of the oldest and simplest games, has served as an ideal subject for computerization due to its straightforward rules and minimal computational requirements. (Balagurusamy, 2020) Historically, computer implementations of Rock-Paper-Scissors were basic, often limited to single-player modes against the computer without persistent data storage. These implementations lacked features such as user registration, match history, and leaderboards, which are essential for fostering competitive spirit and long-term engagement. With the advent of modern programming languages and database technologies, developers began integrating file handling and database management systems to store user profiles, game results, and historical data. This transition has improved the scalability and user experience of such games. (Prata, 2014; Microsoft, n.d.)

Current challenges in digital game implementations include data security, user friendliness, and feature richness. Many existing systems do not support multiplayer functionality, which limits social interaction. Additionally, traditional game systems often do not include comprehensive dashboards or real-time leaderboards, which are critical for motivating players. As gaming and computing developed, many traditional games were adapted into digital versions. Early computer versions of simple games were usually built to demonstrate basic logic and user interaction. Rock Paper Scissors became especially useful for programming practice because it requires only a few rules, yet it still needs conditional decision-making and random number generation. In this way, it became a common beginner-level project in programming education. The digital version also removed the need for physical play and allowed the game to be used in classrooms, labs, and self-learning environments.

Many existing systems do not support multiplayer functionality, which limits social interaction. Additionally, traditional game systems often do not include comprehensive dashboards or real-time leaderboards, which are critical for motivating players. (GeeksforGeeks, n.d.) This project aims to address these gaps by implementing a multi-mode Rock-Paper-Scissors game with robust data management, multiplayer capabilities, and an intuitive user interface. The motivation stems from the need to create an engaging online game platform that is accessible to users with varying levels of technical expertise, and to explore how traditional games can be adapted to modern digital environments. The development process involves analysing existing systems, identifying missing features, and designing an integrated system that combines game logic with user management, data persistence, and interactive interfaces. (Balagurusamy, 2020; Kernighan & Ritchie, 1988)

The project is also suitable for a local academic setting because it can be implemented with simple tools and tested on standard computers without needing internet access. This makes it accessible to students and teachers while still showing important concepts such as modular programming, data persistence, and user interaction. In short, the background of the project shows a clear evolution from a traditional hand game to a digital system that supports storage, tracking, and repeated use. (Pokhara University, n.d.)

5 .Requirement Document

A project cannot be successfully implemented without proper requirement analysis. The requirement document identifies the needs of stakeholders and ensures that the developed system meets user expectations. The Rock Paper Scissors Game was developed to provide a digital platform for playing the traditional game while incorporating user authentication, score tracking, and file management features.

The requirement gathering process involved identifying the needs of players, developers, and supervisors. These requirements were categorized into business requirements, user requirements, functional requirements, and non-functional requirements.

5.1 Requirement Matrix

SN.	Required modules, system, and features.	Description for the modules	Priority (High, Moderate, low)	Remarks
1.	Registration system and Login System	Allows new users to create an account with username and password and Authenticates registered users before accessing the game.	High	Essential for first-time users and security feature .
2.	Password Validation	Verifies user credentials during login	High	Prevents unauthorized access.

3.	Main Dashboard	Provides navigation to all game functions.	High	Central control module
4.	Single Player Mode	Allows users to play against the computer.	High	Core functionality
5.	Random Choice Generator	Generates computer moves using random numbers.	High	Required for gameplay
6.	Multiplayer Mode	Allows two users to play on the same system.	High	Additional functionality
7.	Winner Determination Module and Score Management System	Determines winner according to game rules and updates scores	High	Main game logic and Performance tracking.
8.	File Handling System and Scoreboard Display	Stores user accounts and scores in files and Shows player records and scores.	High	Data persistence and User convenience
9.	Input Validation	Prevents invalid menu selections and entries.	High	Error handling.

10.	Logout System and Exit System	Allows users to safely log out and Closes the application safely.	High	Security enhancement and final operation.

5.2 Business Requirements

1.Digitalize the traditional Rock Paper Scissors game

2.Reduce manual score keeping

3. Maintain player records securely.

4.Provide an interactive gaming experience

5.3 User Requirements

1.User should be able to register.

2.User should be able to login.

3.User should be able to play against computer.

4.User should be able to play multiplayer mode.

5.User should be able to view scores.

5.4 Functional Requirements

1.Registration functionality.

2.Login authentication.

3. Gameplay processing.

4. Winner determination.

5.Score calculation.

6.File storage and retrieval.

5.5 Non-Functional Requirements

1.Performance Reliability

2.Security

3. Usability

4.Maintainability

5.Portability

6 .System Design

System Design represents the structure and operation of the Rock Paper Scissors Game. The design phase converts system requirements into a blueprint for development.

6.1 Data Flow Diagram (DFD)

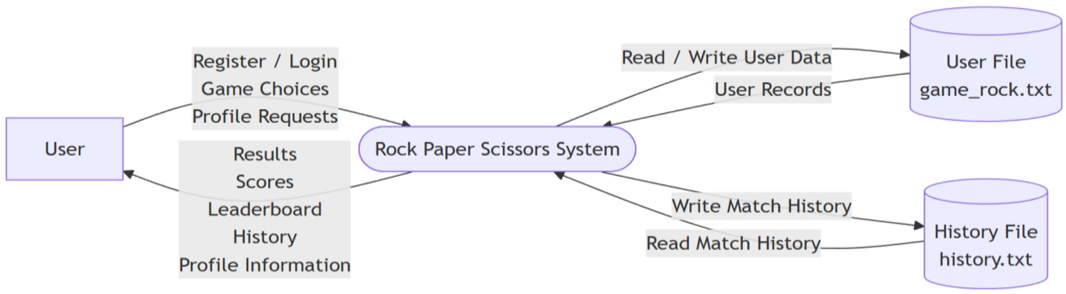

Figure 6.1 : Level 0 DFD of Rock, Paper, Scissor

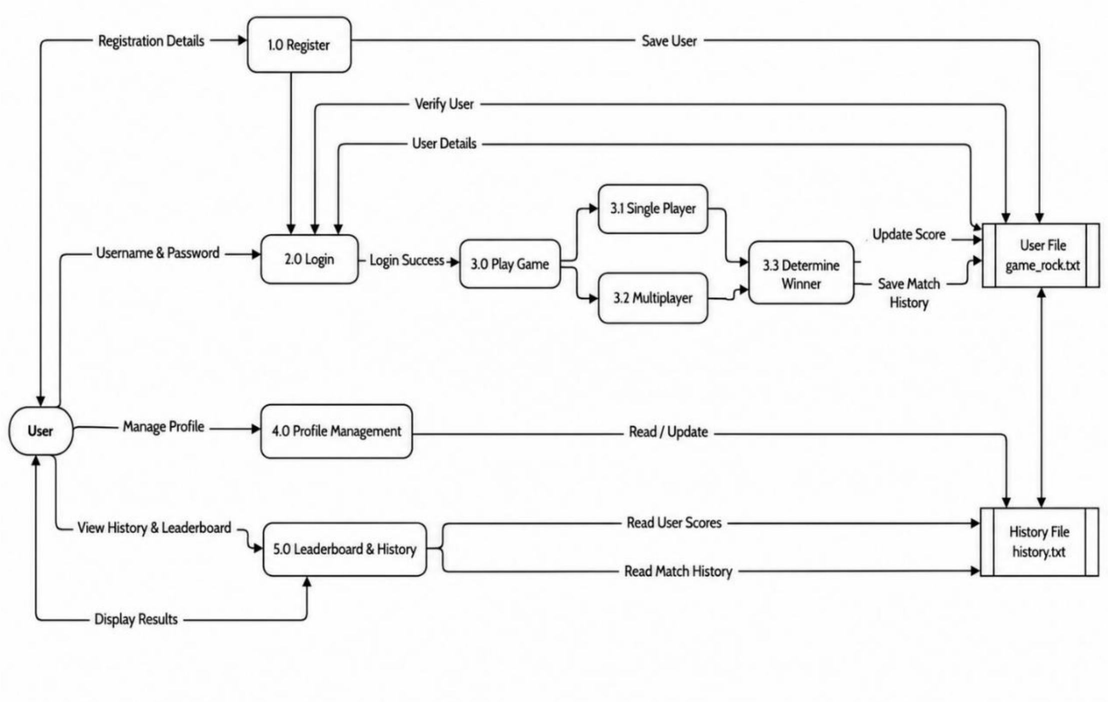

Figure 6.2 : Level 1 DFD of Rock, Paper, Scissor

6.2 Flowchart

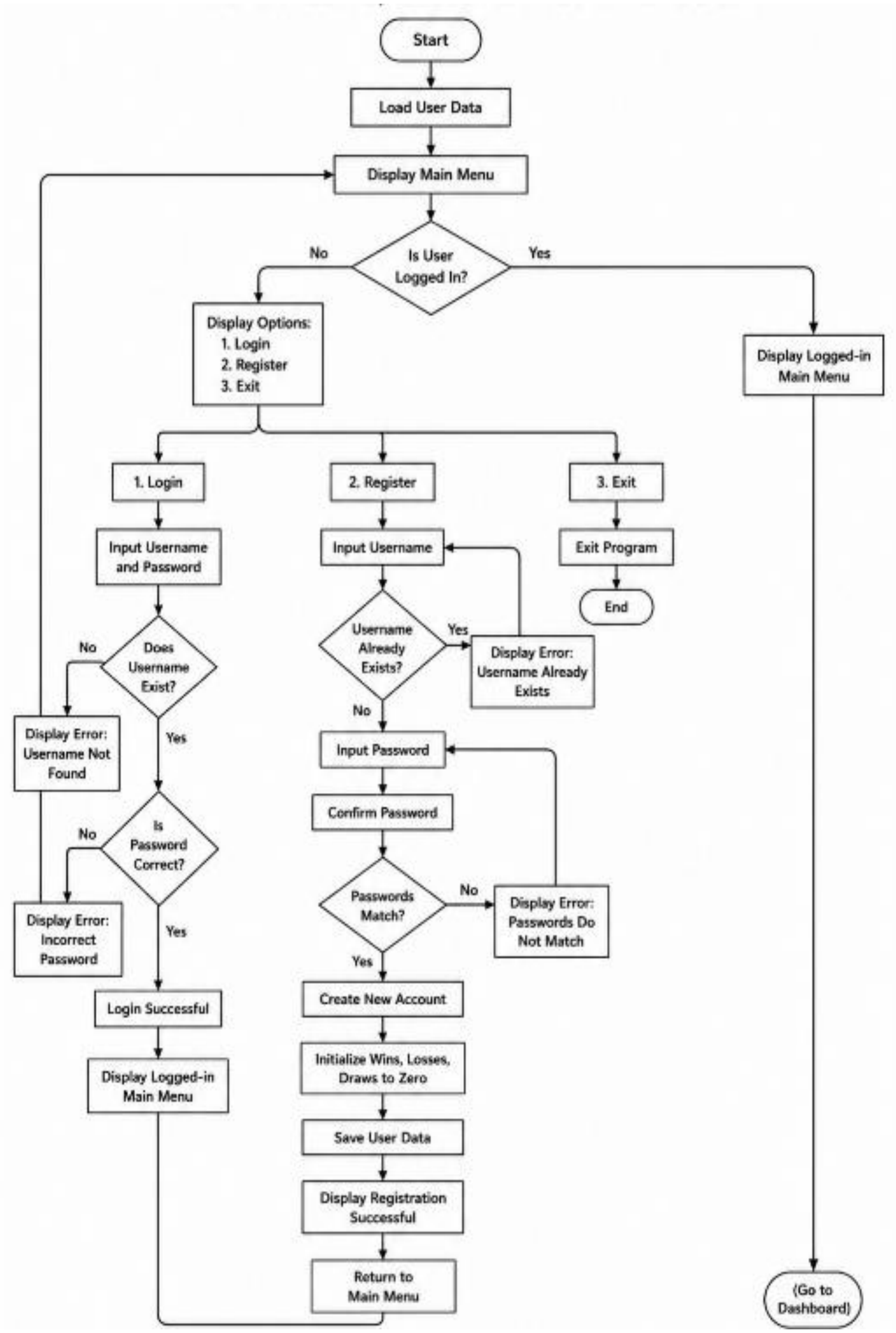

Figure 6.3 : Flowchart for main menu, login and registration

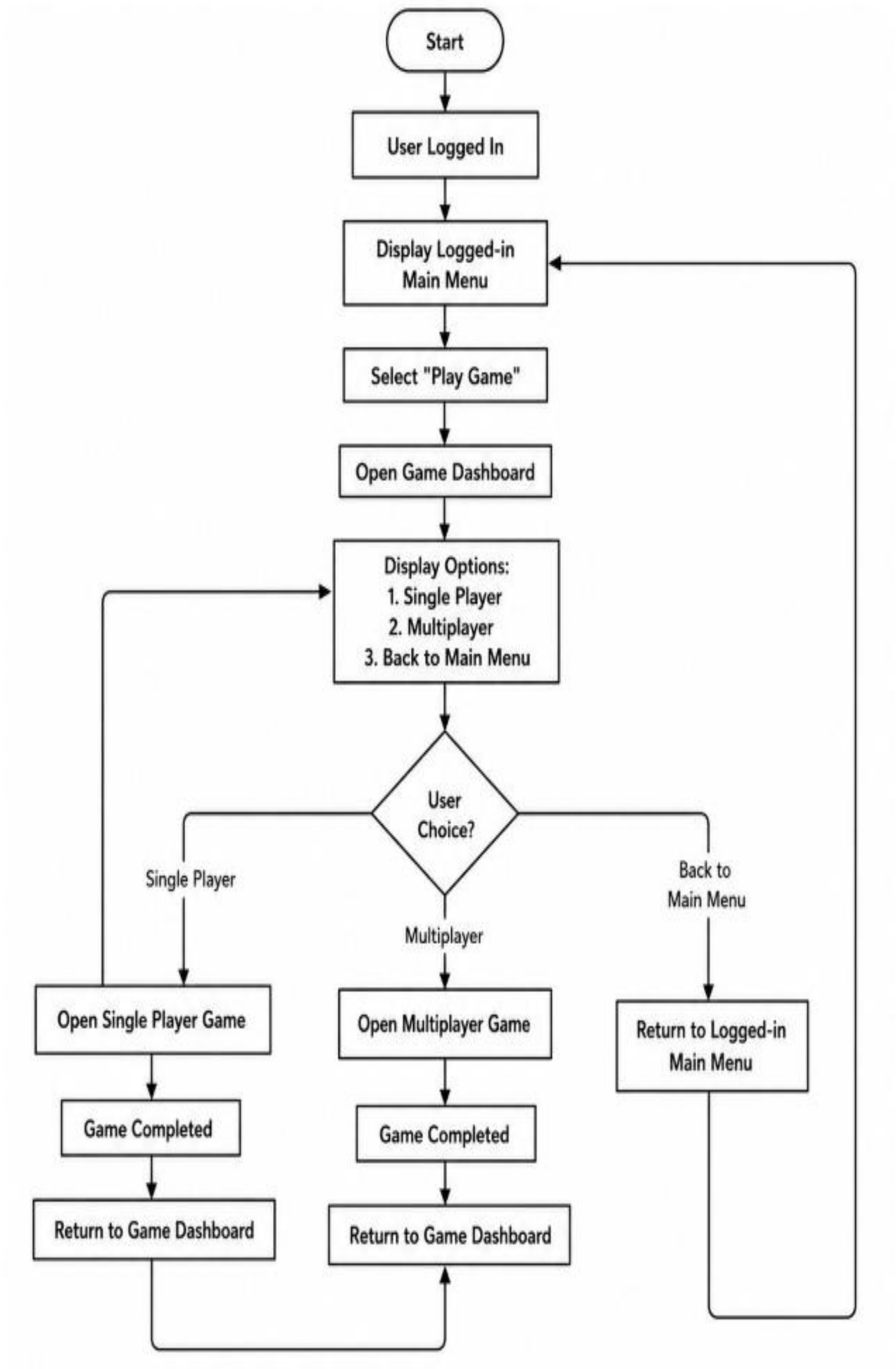

Figure 6.4 : Flowchart for Dashboard

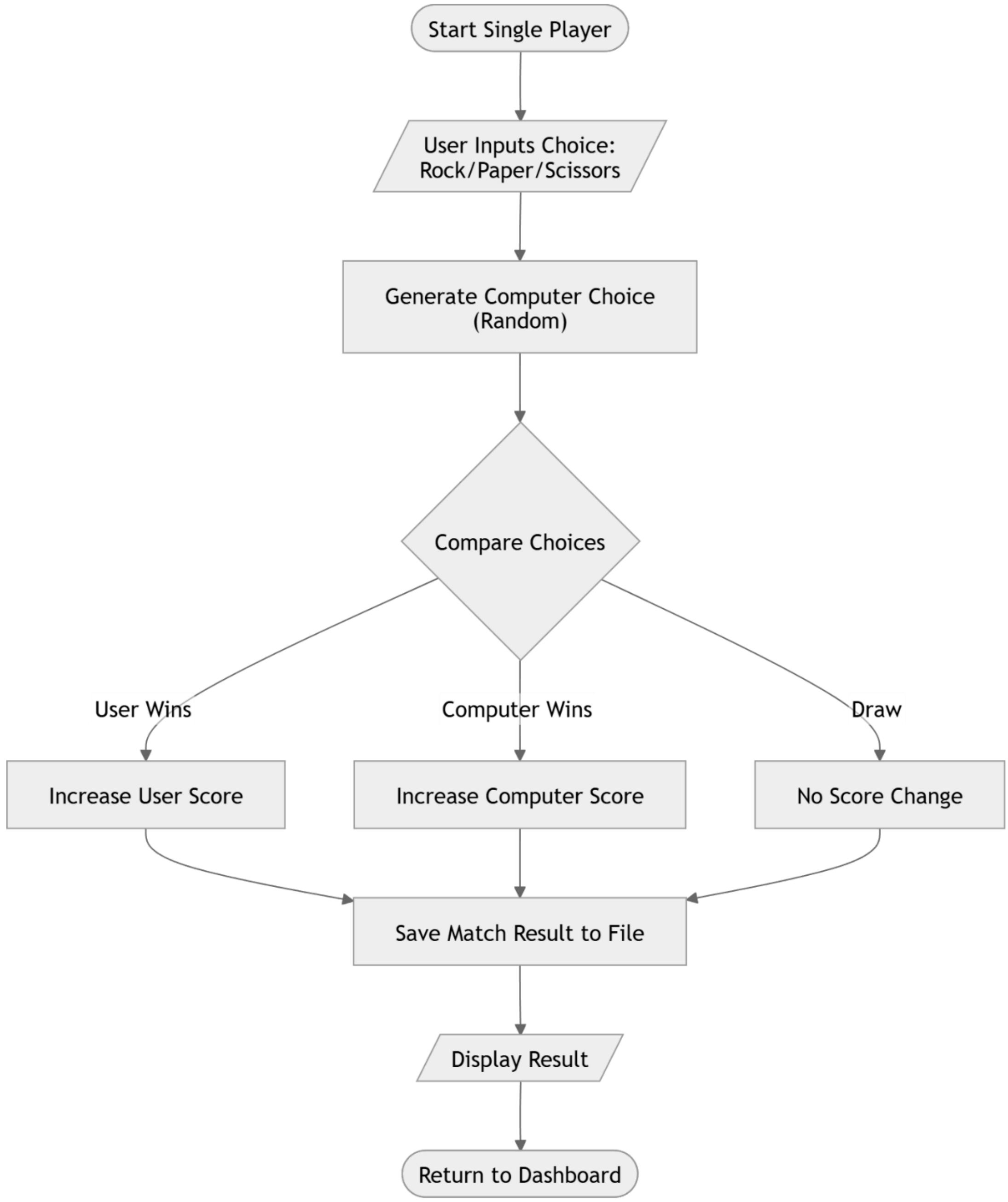

Figure 6.5 : Flowchart for Single Player Gameplay

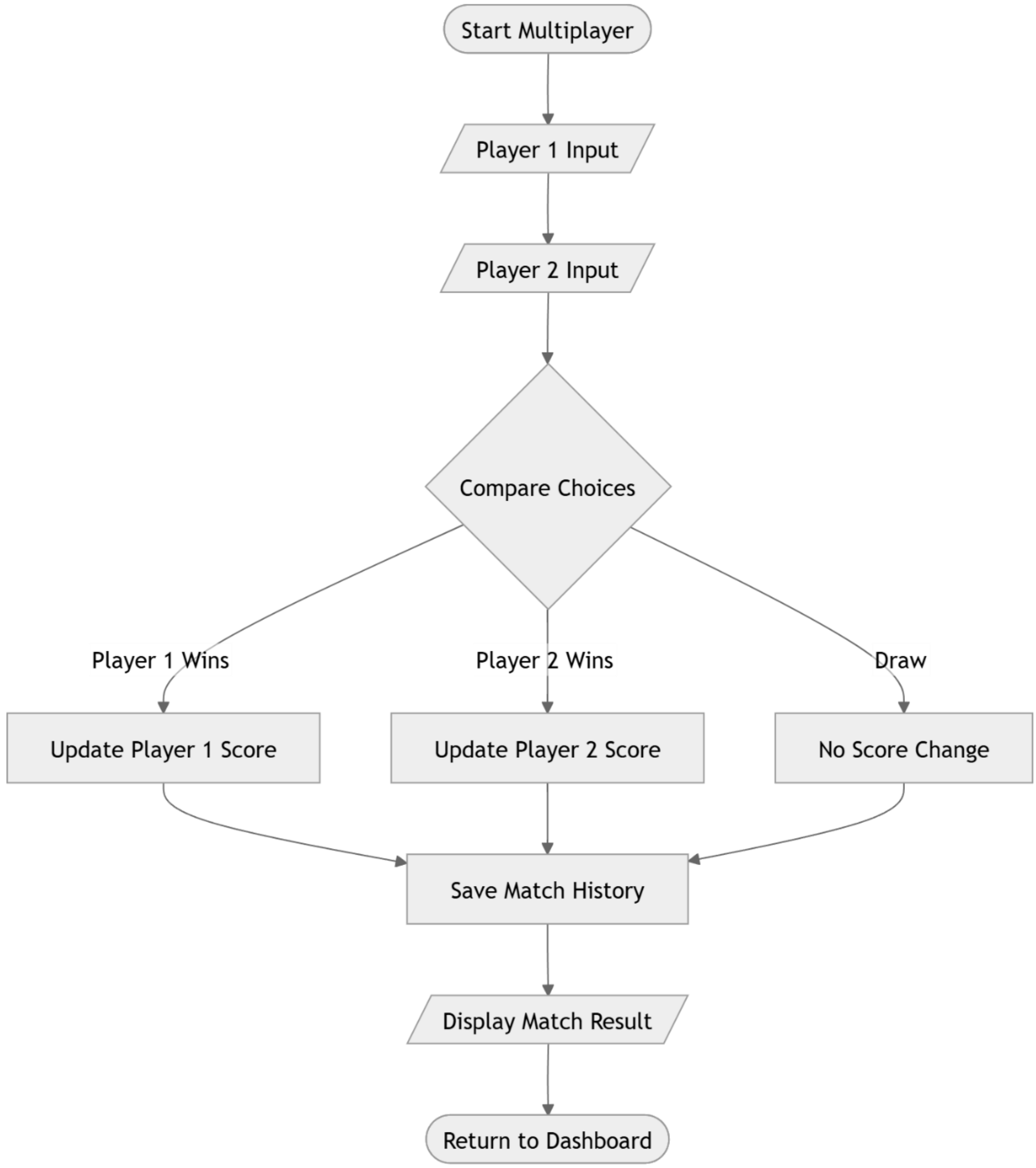

Figure 6.6 : Flowchart for Multiplayer Gameplay

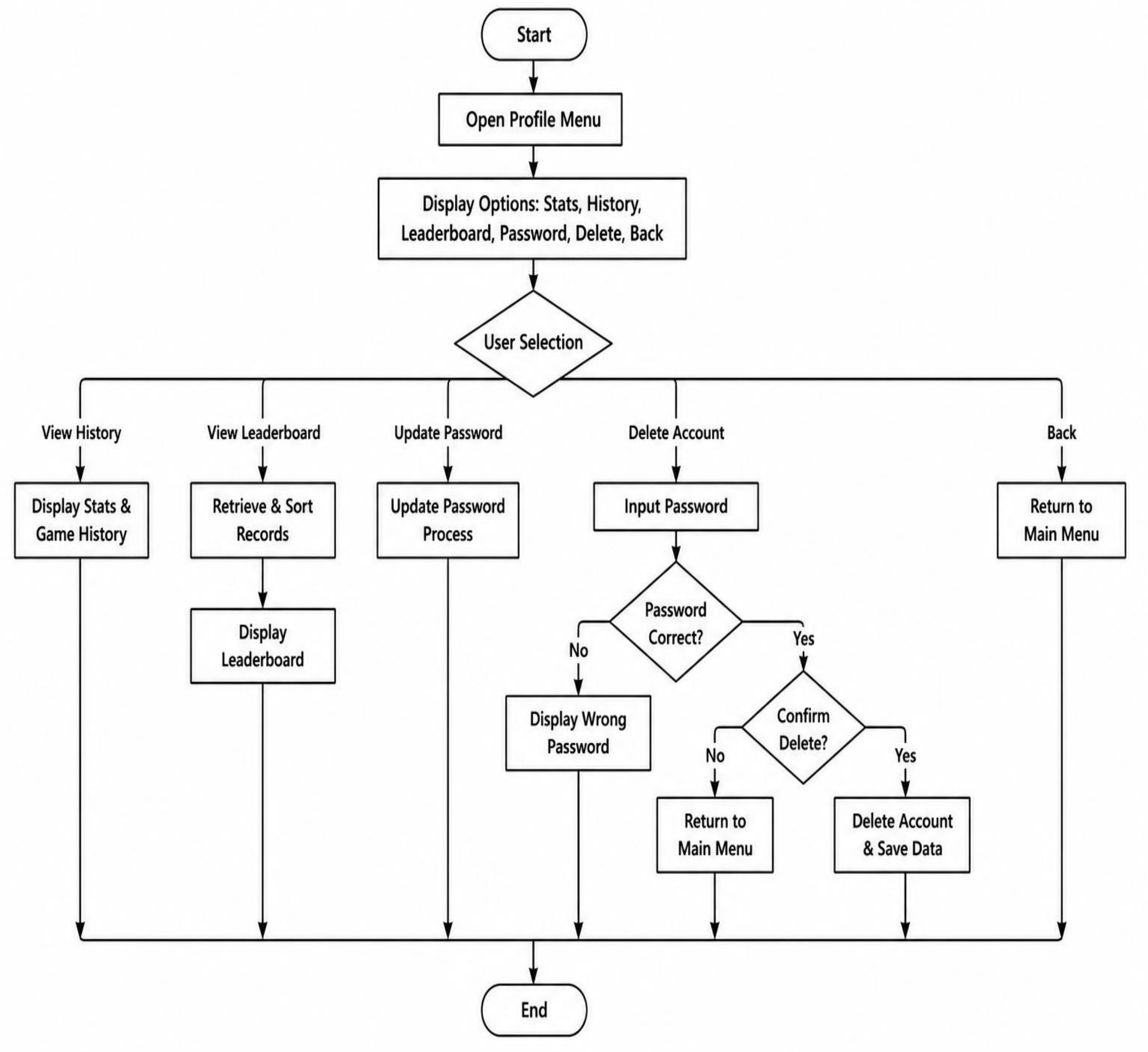

Figure 6.7 : Flowchart for History, Leader board and Delete

6.3 Use Case Diagram

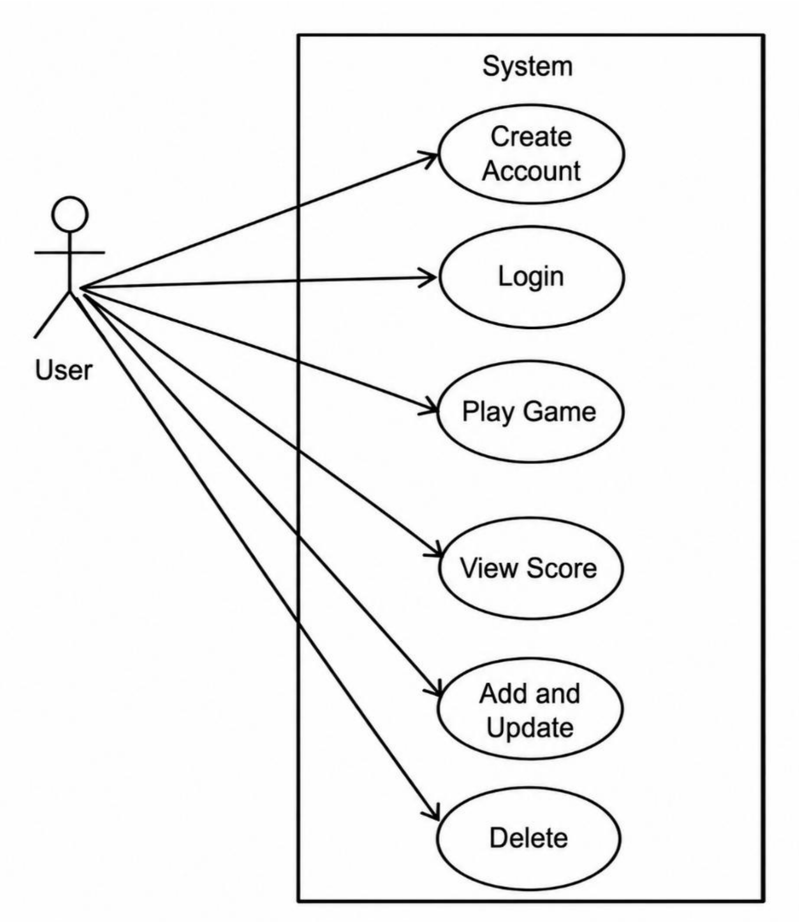

Figure 6.8 : Use Case diagram

7 . Development

7.1 Waterfall Model

The waterfall model was chosen for the development of the Rock Paper Scissor game because it is suitable for small and well-defined projects. The project requirements and features are clearly defined before the development process which made the waterfall approach effective and easy to manage. Unlike complex software projects, this project follows a simple and structured process. Each phase such as requirement analysis, system design, implementation, testing, and maintenance was completed step by step in sequence which helped in proper planning, coding, testing, and documentation.
The Waterfall Model was preferred over other models because:

1. The project had fixed and clearly defined requirements.
2. The system was small-scale and easy to manage.
3. Development was completed phase by phase in an organized manner.
4. It was suitable for academic and documentation-based projects.
5. The model provided a clear structure and proper documentation for each stage.

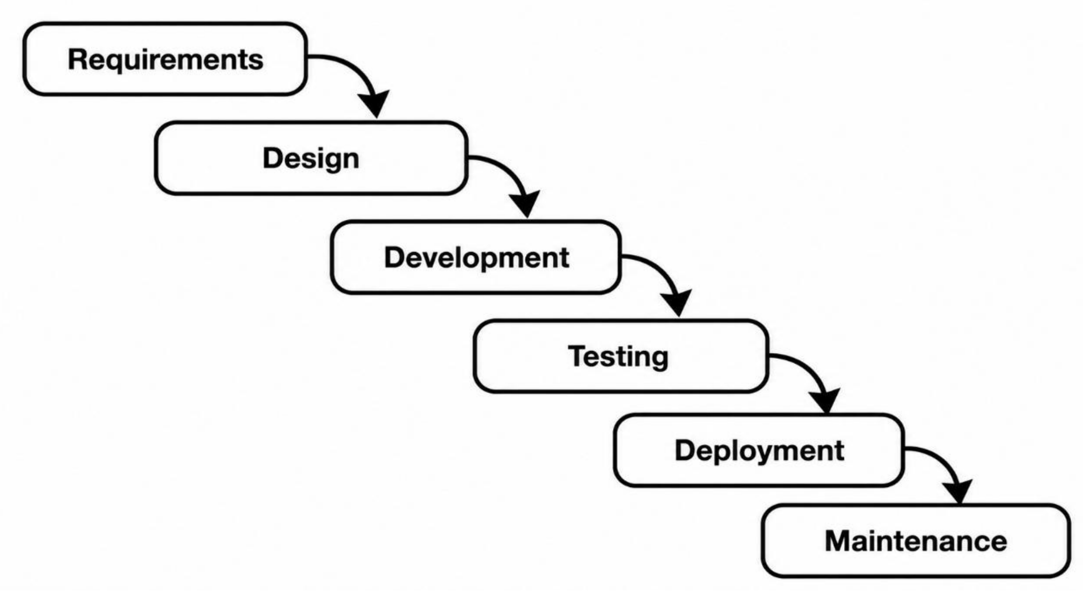

Figure 7.1 : Waterfall Model

Finally, most importantly, the chart of work assignment.

S.N.	Name of the member	Work assigned	Remarks
1.	Divya Sharma	Documentation and coding. • DFD • Problem Identification • Coding • System design.	
2.	Sabina Puwanli	Documentation and coding. • Gantt Chart • Coding • Support in system design. • Feasibility analysis.	

7.2 Gantt chart

The Gantt chart is a graphical representation of the project schedule that shows the sequence and duration of activities involved in the development of the Rock Paper Scissors game. It helps in planning, monitoring, and completing the project within the allocated time frame. Since this project follows the Waterfall Model, the development process is divided into different phases such as requirement analysis, system design, coding, testing, documentation, and final deployment. The chart below illustrates the timeline of each phase from the beginning of the project to the final submission period.

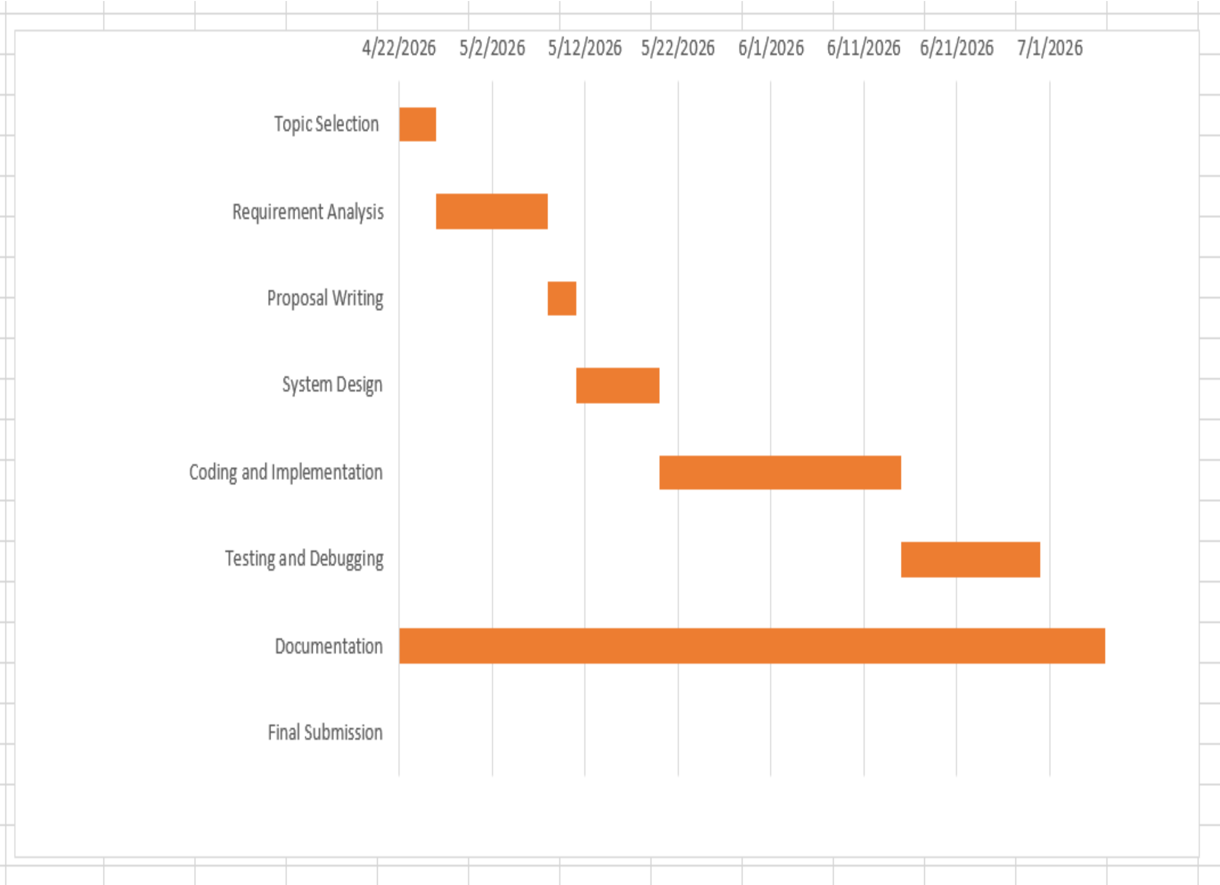

Figure 7.2 : Gantt Chart

8 .Testing

Testing is one of the most important phases of the Software Development Life Cycle (SDLC). It is performed to ensure that the developed software functions correctly according to the specified requirements. Software Testing Life Cycle (STLC) is a systematic process followed to identify defects, verify functionality, and improve software quality before deployment.

The Software Testing Life Cycle (STLC) followed in this project consists of six phases:

1. Requirement Analysis

The project requirements were analyzed to identify the functionalities that needed testing. These include registration, login, gameplay, score management, history, leaderboard, profile update, and account deletion.

2. Test Planning

A test plan was prepared to define the objectives, testing methods, and expected outcomes for every module.

3. Test Case Design

Test cases were designed for both valid and invalid inputs to verify the correctness of each feature.

4. Test Environment Setup

The application was compiled and executed using the Dev-C++ compiler on Windows. The binary data file (**game_rock.txt**) was created to store user information.

5. Test Execution

Each module was executed independently and then integrated to verify complete system functionality.

6. Test Closure

After successful execution of all test cases, the application was approved for deployment.

8.1 Test Case 1 (User Registration)

Objective	To validate successful user registration.
Action	Enter a unique username, password, and confirm password.
Expected Result	User account should be created successfully.
Actual Result	User account created successfully and stored in game_rock.txt.
Conclusion	Test was successful.
Test Executed By	Divya Sharma
Test Reviewed By	Sabina Puwanli

8.2 Test Case 2 (User Login)

Objective	To validate login using correct credentials.
Action	Enter registered username and password.
Expected Result	User should successfully access the Game Dashboard.
Actual Result	Login completed successfully.
Conclusion	Test was successful.
Test Executed By	Sabina Puwanli
Test Reviewed By	Divya Sharma

8.3 Test Case 3 (Single Player Gameplay)

Objective	To verify the Single Player module.
Action	Login, select Single Player, and choose Rock, Paper, or Scissors.

Expected Result	Computer generates a random choice, determines the winner, and updates the score.
Actual Result	Winner displayed correctly and score updated successfully.
Conclusion	Test was successful.
Test Executed By	Divya Sharma
Test Reviewed By	Sabina Puwanli

8.4 Test Case 4 (Multiplayer Gameplay)

Objective	To verify multiplayer gameplay.
Action	Login, select Multiplayer, enter opponent username, and both players make their choices.
Expected Result	Winner should be determined correctly and both players' scores updated.
Actual Result	Match result displayed correctly and scores updated successfully.
Conclusion	Test was successful.
Test Executed By	Divya Sharma
Test Reviewed By	Sabina Puwanli

8.5 Test Case 5 (Leaderboard)

Objective	To verify leaderboard generation.
Action	Play multiple games and open the Leaderboard option.

Expected Result	Users should be displayed in descending order according to total wins.
Actual Result	Leaderboard displayed correctly after sorting users by wins.
Conclusion	Test was successful.
Test Executed By	Sabina Puwanli
Test Reviewed By	Divya Sharma

8.6 Requirement Traceability Matrix (RTM)

S.N.	Requirement Description	Test Case ID	Status	Remarks
101	User Registration	TC-001	Pass	User account created successfully
102	Login Authentication	TC-002, TC-003	Pass	Invalid credentials handled correctly
103	Single Player Mode	TC-004	Pass	Winner calculated correctly
104	Multiplayer Mode	TC-005	Pass	Both player scores updated correctly
105	Score Management	TC-006	Pass	Wins, losses and draws updated
106	Match History	TC-007	Pass	History stored successfully
107	Leaderboard	TC-008	Pass	Users sorted according to wins
108	Profile Update	TC-009	Pass	Password updated successfully

109	Delete Account	TC-010	Pass	Account removed successfully
110	File Handling	TC-011	Pass	User data stored and retrieved correctly

8.7 User Acceptance Testing (UAT)

No.	Acceptance requirement (USR)	Critical (DEV)		Test Results (DEV)		Comments (USR)
		Yes	No	Accept	Reject	
1.	User can register a new account successfully.	Affirm.		Affirm		Successfully accepted.
2.	Registered users can log in using valid credentials.	Affirm.		Affirm.		Login works correctly.
3.	Single Player mode functions correctly.	Affirm.		Affirm.		Winner determined accurately.
4.	Multiplayer mode updates scores correctly.	Affirm.		Affirm.		Gameplay works correctly
5.	Leaderboard displays rankings correctly.	Affirm.		Affirm.		Users sorted by total wins.
6.	Match history is stored successfully.	Affirm.		Affirm.		History displayed correctly.
7.	User data remains available after restarting the application.	Affirm.		Affirm.		File handling works correctly.

9 .Project Results

The Rock Paper Scissors Game was developed to provide a simple, interactive, and user-friendly digital version of the traditional game using C programming language. The project was designed to implement important programming concepts such as user authentication, file handling, game logic, score management, and modular programming. The objectives identified during the planning phase were successfully achieved through the developed system.

The problems solved by this project are:

1. Manual score keeping is eliminated as the system automatically records wins, losses, and draws.
2. User accounts can be securely managed through registration and login functionality.
3. Player records and match history are stored permanently using file handling, allowing data to remain available even after the program is closed.
4. Players can view their game statistics and leaderboard without maintaining records manually.
5. Both single-player and multiplayer gameplay are available within the same application.

After studying the limitations of the traditional Rock Paper Scissors game, we identified the need for a computerized system that could manage player information and game results efficiently. The developed system addresses these limitations by providing the following benefits:

1. Fast and easy user registration and login.
2. Automatic winner determination based on the game rules.
3. Permanent storage of player information and match history using files.
4. Easy access to player statistics, history, and leaderboard.
5. A simple menu-driven interface that is easy for users to understand and operate.

During the development of this project, we gained practical knowledge of C programming concepts such as structures, functions, arrays, conditional statements, loops, file handling, and modular programming. We also learned how to design, develop, test, and document a complete software project.

Working as a team helped us improve our communication, coordination, and problem-solving skills. We learned how to divide tasks, support each other during development, and solve programming challenges together. Overall, this project provided valuable practical experience and increased our confidence in developing real-world software applications using C programming.

10 .Future Enhancements

The current Rock Paper Scissors Game successfully implements user registration, login, gameplay, score management, leaderboard, match history, and file handling. However, the project can be further improved by adding more advanced features in the future. Some possible future enhancements are:

1. Develop a Graphical User Interface (GUI) to make the game more attractive and user-friendly.
2. Replace file handling with a database management system to improve data storage, security, and scalability.
3. Add online multiplayer functionality using socket programming so that players can compete over a network.
4. Introduce an AI-based computer opponent with different difficulty levels.
5. Implement tournament mode with multiple rounds and automatic tournament winner selection.
6. Add player profiles with avatars, achievements, and badges.
7. Include sound effects, animations, and improved visual design to enhance the gaming experience.
8. Develop a mobile or web-based version of the game so that users can play from different devices.
9. Improve the leaderboard by adding player rankings, win percentages, and detailed game statistics.
10. Strengthen the security of user accounts by implementing password encryption and improved authentication methods.

These enhancements would make the system more interactive, secure, and suitable for a large number of users while providing a more enjoyable gaming experience.

11 . Conclusion

The Rock Paper Scissors Game was successfully developed using the C programming language as a console-based application. The project achieved its main objective of creating a digital version of the traditional game while applying important programming concepts such as functions, structures, arrays, conditional statements, loops, and file handling.

The project began by identifying the limitations of the traditional manual game and defining clear objectives and system requirements. Based on these requirements, the system was designed using appropriate diagrams including the Data Flow Diagram (DFD), flowcharts, and use case diagram. The Waterfall Model was followed throughout the development process, allowing each phase of the project to be completed in a systematic manner.

The developed system provides essential features such as user registration, login, single-player mode, multiplayer mode, score management, match history, leaderboard, profile management, and permanent data storage through file handling. These features make the game more organized, interactive, and user-friendly compared to the traditional version.

The project also provided valuable practical experience in software development. It improved our understanding of program design, debugging, file management, testing, documentation, and teamwork. Throughout the project, we learned how to convert theoretical knowledge into a working software application while solving real programming challenges.

Overall, the Rock Paper Scissors Game successfully meets its objectives and demonstrates the practical application of C programming concepts. The project serves as a strong foundation for developing more advanced gaming applications in the future by incorporating graphical interfaces, networking, and database technologies.

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13 . Annexures

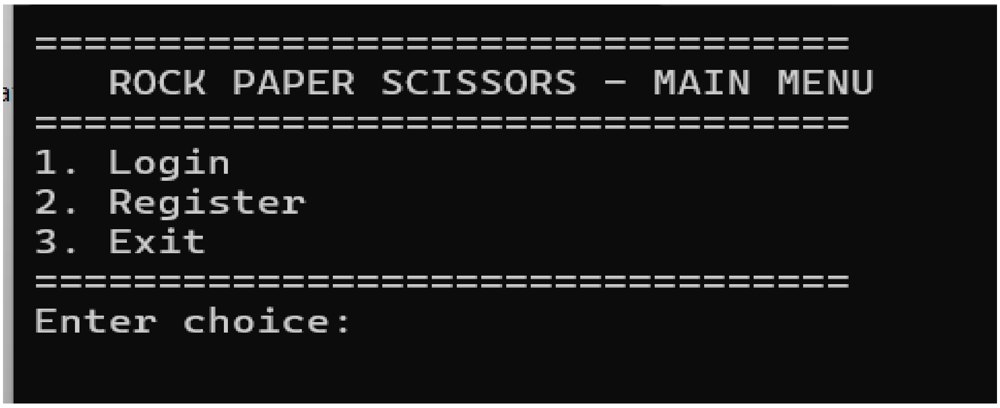

Figure 13.1 : Main Menu

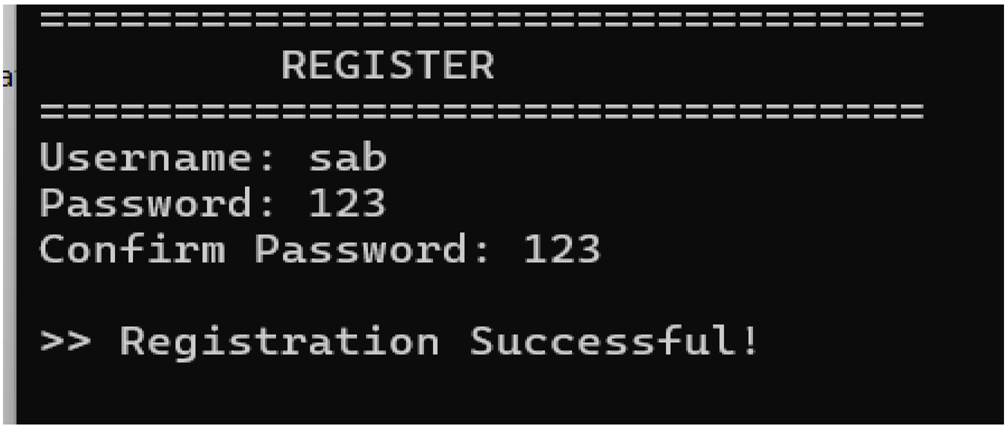

Figure 13.2 : User Registration

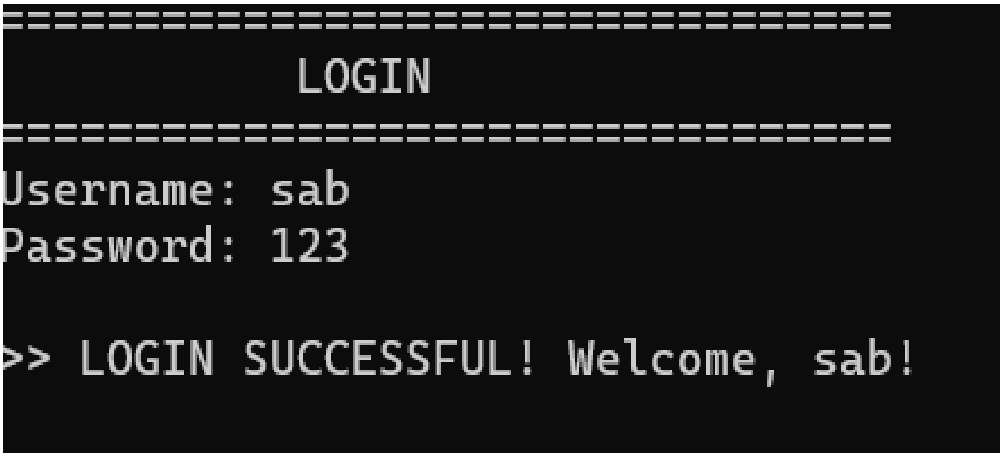

Figure 13.3 : User Login

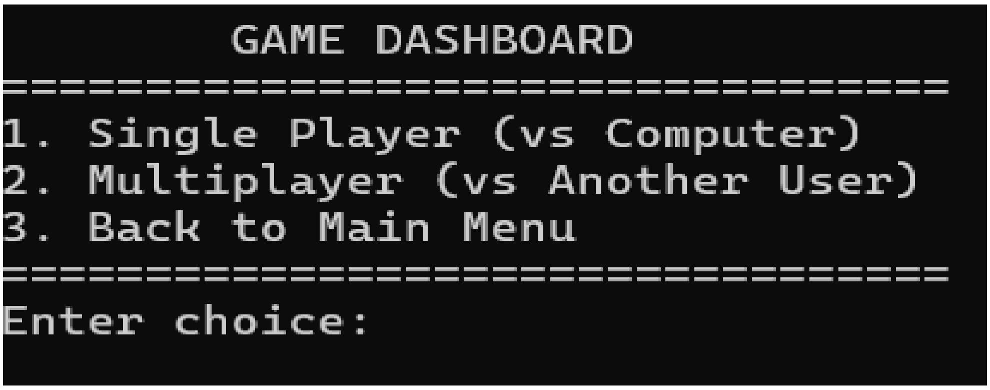

Figure 13.4 : Game Dashboard

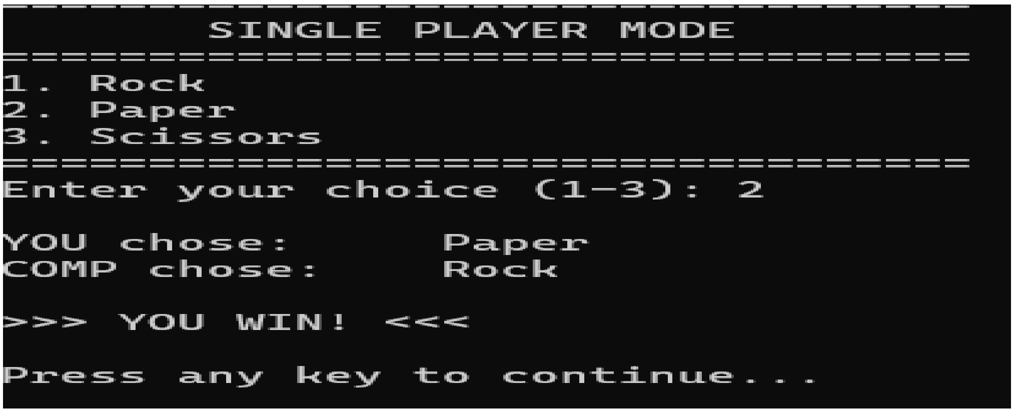

Figure 13.5 : Single player Gameplay

Figure 13.6 : Profile Menu

```
==================================
          LEADERBOARD
==================================
Name                       Wins    Loss    Draw
sab                           2       0       0
malika22                      1       0       0
ded                           0       1       0
```

Figure 13.7 : Leader board

```
Username: sab
Wins: 2 | Losses: 0 | Draws: 0
History: Dual: Scissors vs Paper [W] | Solo: Paper vs Rock [W] | New User
```

Figure 13.8 : View History